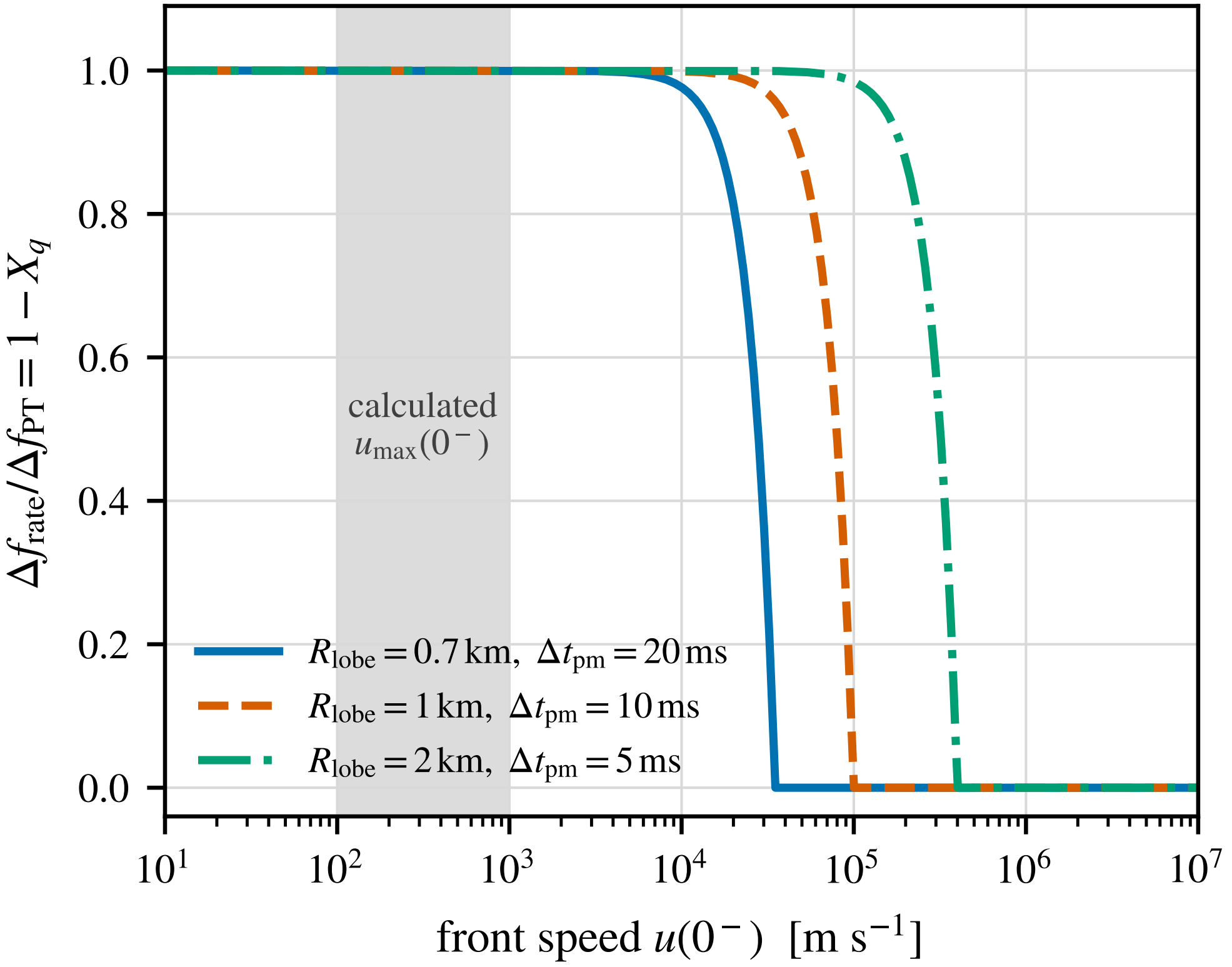
1.0
0.8
0.6
0.4
0.2
0.0
$\Delta f_{\rm rate}/\Delta f_{\rm PT} \equiv 1 - X_q$
calculated
$u_{\rm max}(0^-)$
$R_{\rm lobe} = 0.7\,{\rm km},\ \Delta t_{\rm pm} = 20\,{\rm ms}$
$R_{\rm lobe} = 1\,{\rm km},\ \Delta t_{\rm pm} = 10\,{\rm ms}$
$R_{\rm lobe} = 2\,{\rm km},\ \Delta t_{\rm pm} = 5\,{\rm ms}$
10^1
10^2
10^3
10^4
10^5
10^6
10^7
front speed $u(0^-)$ [m s^{-1}]